Tutorial Sheet-1

Indian Institute of Technology Jodhpur
MAL1020 (Mathematics-II)
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1. In the following, a set V, a field $\mathbb{F}$, which is either $\mathbb{R}$ or $\mathbb{C}$, and the operations of addition and scalar multiplication are given. Check whether V is a vector space over $\mathbb{F}$ with these operations.
   (a). With addition and scalar multiplication as in $\mathbb{R}^2$, let $V = \big\{(a, b) \in \mathbb{R}^2 : 2a + 3b = 0\big\}$ and $\mathbb{F} = \mathbb{R}$.
   (b). Let $V = \mathbb{R}^2$ and $\mathbb{F} = \mathbb{R}$. with addition as in $\mathbb{R}^2$, and scalar multiplication as given by the : for $(a, b) \in V, \alpha \in \mathbb{R}$, $\alpha(a, b) := (a, 0)$.
   (c). Let $V = \{x \in \mathbb{R} : x > 0\}, \mathbb{F} = \mathbb{R}$, and for $x, y \in V, \alpha \in \mathbb{R}$, define the operations as follows: $x + y := xy, \alpha x := x^{\alpha}$. The known operations on the right define the operations on the left.

2. Let V_1 and V_2 be two vector spaces over the same field $\mathbb{F}$ with respect to the same operations and both have the same null vector. Then will their union be a vector space over $\mathbb{F}$? Justify your answer. What do you think about their intersection?

3. Let $V = \{A \in M_{n\times n} : \operatorname{tr}(A) = 0\}$ be a subset of $M_{n\times n}$, the set of all $n \times n$ matrices with real entries. Verify whether or not V is a vector space over $\mathbb{R}$.

4. Examine if the set S is a subspace of $\mathbb{R}^3$
   (a). $S = \big\{(x, y, z) \in \mathbb{R}^3 : x = 0\big\}$,
   (b). $S = \big\{(x, y, z) \in \mathbb{R}^3 : xy = z\big\}$,
   (c). $S = \big\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\big\}$,
   (d). $S = \big\{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0, 2x - y + z = 0\big\}$.

5. Examine if the set S is a subspace of the Vector space $\mathbb{R}_{2\times 2}$, where
   (a). $S = \left\{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}_{2\times 2} : a + b = 0\right\}$,
   (b). $S = \left\{\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}_{2\times 2} : \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = 0\right\}$,

6. Determine which of the following sets are bases for $\mathbb{R}^3$.
   (a). $\{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$
   (b). $\{(2, -4, 1), (0, 3, -1), (6, 0, -1)\}$
   (c). $\{(1, 2, -1), (1, 0, 2), (2, 1, 1)\}$
   (d). $\{(-1, 3, 1), (2, -4, -3), (-3, 8, 2)\}$
   (e). $\{(1, -3, -2), (-3, 1, 3), (-2, -10, -2)\}$

7. For what real values of k does the set S form a basis of $\mathbb{R}^3$?
   (a). $S = \{(k, 1, k), (0, k, 1), (1, 1, 1)\}$;
   (b). $S = \{(k, 0, 1), (1, k + 1, 1), (1, 1, 1)\}$.

8. Find the dimension of the subspace S of $\mathbb{R}^3$ defined by,
   (a). $S = \{(x, y, z) \in \mathbb{R}^3 : 2x + y - z = 0\}$;
   (b). $S = \{(x, y, z) \in \mathbb{R}^3 : x + 2y = z, 2x + 3z = y\}$.

9. Find a basis and determine the dimension of the following subspace S of the vector space $\mathbb{R}_{2\times 2}$.
   (a). $S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}_{2\times 2} : a + b = 0 \right\}$,
   (b). $S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}_{2\times 2} : a = d = 0 \right\}$,

10. Determine whether the following sets are linearly dependent or linearly independent.
   (a). $\left\{ \begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}, \begin{pmatrix} -2 & 6 \\ 4 & -8 \end{pmatrix} \right\}$ in $\mathrm{M}_{2\times 2}(\mathbb{R})$
   (b). $\left\{ \begin{pmatrix} 1 & -2 \\ -1 & 4 \end{pmatrix}, \begin{pmatrix} -1 & 1 \\ 2 & -4 \end{pmatrix} \right\}$ in $\mathrm{M}_{2\times 2}(\mathbb{R})$
   (c). $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ in $\mathrm{P}_3(\mathbb{R})$
   (d). $\{(1,2,1), (2,1,1)\}$ in $\mathbb{R}^2(\mathbb{R})$

   If they are linearly independent, extend them up to the basis of the corresponding vector space.